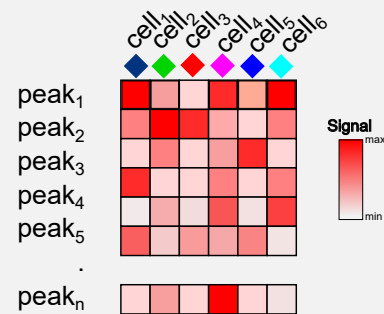


Step 1 - Genome-wide enrichment

Identify globally enriched cell types

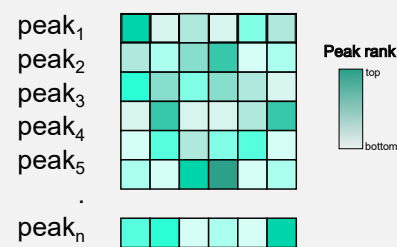
Input 1: snATAC data

- Peak X cell signal matrix



Process snATAC matrix

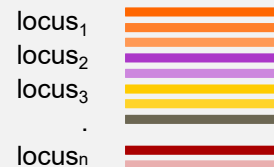
- Noise filter
- Quantile normalisation
- Rank peaks by cell-type specificity score



+

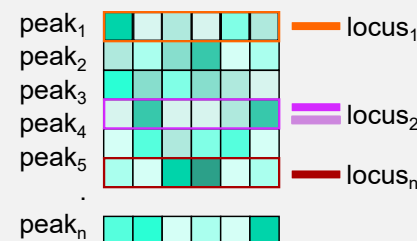
Input 2: GWAS variants

- e.g. Fine-mapped credible sets, high LD proxies



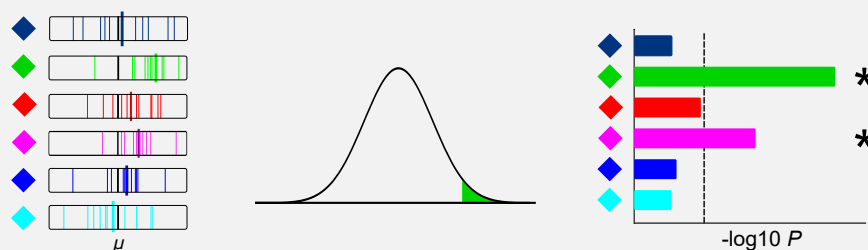
Annotate matrix

- Intersect processed peak matrix with GWAS variants



Analysis

- Compute variant peak mean ranks and significance



Intermediate output:
Globally enriched cell types

Step 2 - Sub-genomewide annotation

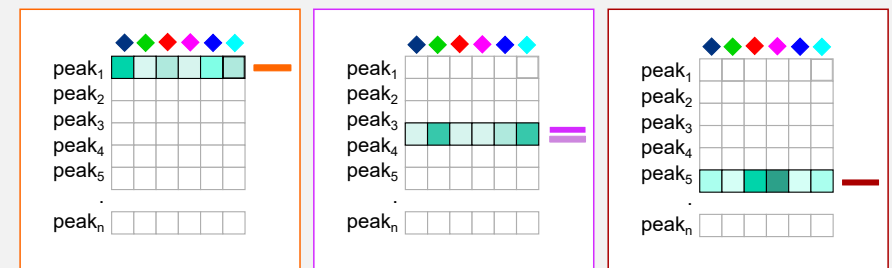
Quantify cell type-specificity at the locus level

Input:

GWAS-annotated peak matrix, ranked by specificity score



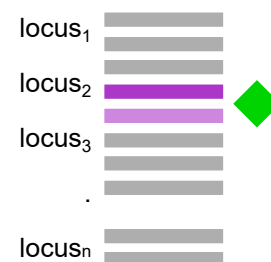
Extract ATAC peaks intersecting GWAS loci



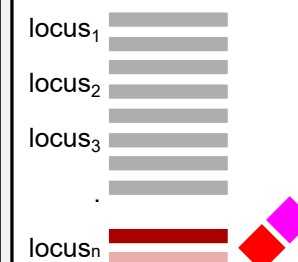
Classify

Cell-type specific peaks defined by rank in top decile

Core-enriched loci



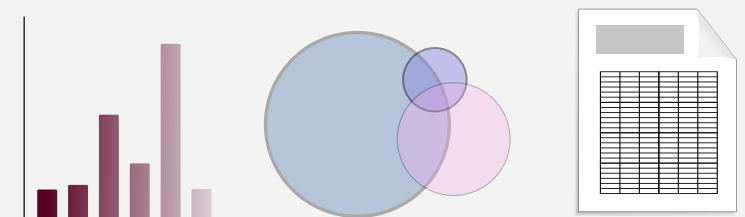
Composite loci



Residual loci



Summarise



Final output:
Locus-level annotation of prioritised driver cell types